

Entangled states

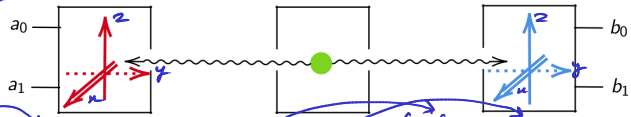
$\rho \equiv \frac{1}{2} \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ in Basis = $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$

$\rho = 1/4 X \otimes X$

$= \frac{1}{2} \left[|01\rangle\langle 01| - |01\rangle\langle 10| - |10\rangle\langle 01| + |10\rangle\langle 10| \right]$

$|01\rangle\langle 01| = |0\rangle\langle 0| \otimes |1\rangle\langle 1| = \left(\frac{1+Z}{2}\right) \otimes \left(\frac{1-Z}{2}\right)$

$|01\rangle\langle 10| = |0\rangle\langle 1| \otimes |1\rangle\langle 0| = \left(\frac{X+iY}{2}\right) \otimes \left(\frac{X-iY}{2}\right)$



$(\hat{n} \cdot \vec{\sigma})(\hat{m} \cdot \vec{\sigma}) = (\hat{n} \cdot \hat{m}) I + i(\hat{n} \times \hat{m}) \cdot \vec{\sigma}$

Entangled state: $|\psi\rangle = \frac{1}{\sqrt{2}} (|01\rangle_{ab} - |10\rangle_{ab})$, $\rho = |\psi\rangle\langle\psi| = \frac{1}{4} (II - \underline{XX} - \underline{YY} - \underline{ZZ})$

$C(a,b) = \langle (\hat{a} \cdot \vec{\sigma})(\hat{b} \cdot \vec{\sigma}) \rangle = \text{tr}(\rho (\hat{a} \cdot \vec{\sigma})(\hat{b} \cdot \vec{\sigma})) = \frac{1}{4} \left[\text{tr}(\hat{a} \cdot \vec{\sigma}) \text{tr}(\hat{b} \cdot \vec{\sigma}) \right]$

Shared randomness

$a = b = x$

a-meas	1	0	1	0	1	...
b-meas	0	1	0	1	0	...

PERFECT quantum corr.

$C(a,b) = -\hat{a} \cdot \hat{b} = -1$

$\neq \underbrace{\text{tr}(x \hat{a} \cdot \vec{\sigma})}_{a_x} \underbrace{\text{tr}(x \hat{b} \cdot \vec{\sigma})}_{b_x}$

$a = b = y$

a-meas	1	1	1	0	0	...
b-meas	0	0	0	1	1	...

PERFECT quantum corr.

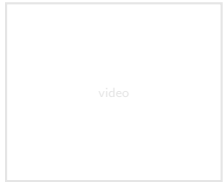
$C(x,y) = -\hat{x} \cdot \hat{y} = 0$

$a = b = z$

a-meas	0	1	1	0	1	...
b-meas	1	0	0	1	0	...

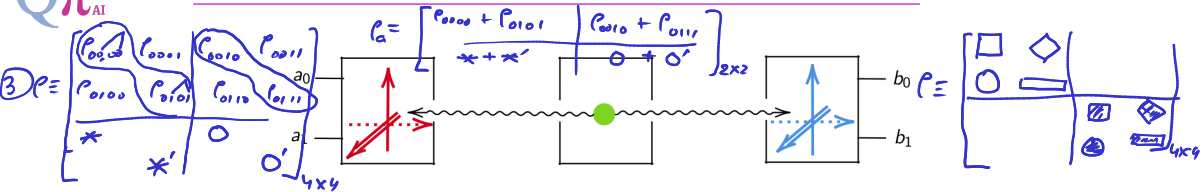
PERFECT quantum corr.

Correlation in more than one measurement-setting and thus stronger than classical



Incomplete measurement, partial trace, reduced state

Basis = $\{ |00\rangle, |01\rangle, |10\rangle, |11\rangle \}$



Entangled state: $|\psi\rangle_{ab} = \frac{1}{\sqrt{2}} (|01\rangle_{ab} - |10\rangle_{ab})$, $\rho_{ab} = |\psi\rangle_{ab}\langle\psi| = \frac{1}{4} \begin{pmatrix} I_a I_b - X_a X_b - Y_a Y_b - Z_a Z_b \end{pmatrix}$

$C(a,b) = -\hat{a} \cdot \hat{b} = -\hat{z} \cdot \hat{x} = 0$

$z = a \neq b = x$

a-meas	1	0	1	0	1	...
b-meas	0	0	1	1	0	...

NO correlation

② $\sum_i \langle i | \rho_a | i \rangle = \rho_a$

Incomplete measurement:

$a = z$

a-meas	1	0	1	0	1	...
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completely random

$Z \otimes I | \psi \rangle = \frac{1}{\sqrt{2}} [|01\rangle + |10\rangle]$

$\langle Z_a \rangle = \text{tr}(Z \otimes I \rho_{ab}) = \text{tr}(Z \rho_a)$

$\langle 4 | Z \otimes I | 4 \rangle = 0$

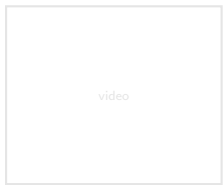
completely mixed

Reduced state by the partial trace:

①

$\rho_a = \text{tr}_b(\rho_{ab}) = \frac{1}{4} (\text{tr}(I) I - \text{tr}(X) X - \text{tr}(Y) Y - \text{tr}(Z) Z) \equiv \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} = \frac{I_a}{2}$

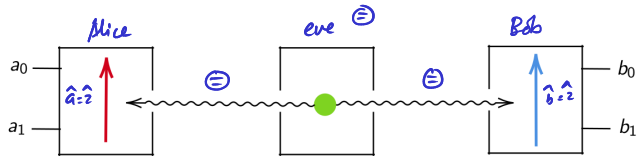
$\rho_b = \text{tr}_a(\rho_{ab}) = \frac{I_b}{2}$



Separable vs Entangled states

$B_2 = \{ |0\rangle, |1\rangle \}$

$0 \leftrightarrow +1$
 $1 \leftrightarrow -1$
 $j \quad z$
 $z = 1 - 2j$
 $z = I - 2P$
 $\hat{a} \cdot \vec{r}$



$\langle e | e^\dagger \rangle = 0$

① Separable states using LOCC: $\rho = \frac{1}{2} (|0\rangle\langle 0| \otimes |1\rangle\langle 1| + |1\rangle\langle 1| \otimes |0\rangle\langle 0|) = \frac{1}{2} [|0\rangle_a |1\rangle_b |e\rangle - |1\rangle_a |0\rangle_b |e^\dagger\rangle]$

$a = b = z$

a-meas	0	1	1	0	1	...
b-meas	1	0	0	1	0	...

PERFECT "classical" correlation

$\rho = \frac{1}{2} \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Hacked $a = b = x$

a-meas	0	0	1	0	1	...
b-meas	0	1	0	0	1	...

NO correlation

$\langle X \otimes X \rangle = \text{tr}(\rho X \otimes X) = \frac{1}{2} (\text{tr}(\rho X \otimes I) \text{tr}(I \otimes X)) = \frac{1}{2} (0 + 0) = 0$

$\langle 10 | X \otimes X | 11 \rangle = \langle 1 | X | 1 \rangle \langle 0 | X | 1 \rangle = 0$

Entangled (non-factorizable) states-vector: $|\psi\rangle_{ab} = \frac{1}{\sqrt{2}} (|0\rangle_a |1\rangle_b - |1\rangle_a |0\rangle_b)$

$a = b = z$

a-meas	0	1	1	0	1	...
b-meas	1	0	0	1	0	...

PERFECT correlation

Not Hacked $a = b = x$

a-meas	1	0	1	0	1	...
b-meas	0	1	0	1	0	...

PERFECT correlation

$X \otimes X |\psi\rangle = \frac{1}{\sqrt{2}} (|1\rangle|0\rangle - |0\rangle|1\rangle) = -|\psi\rangle$

$\langle 4 | X \otimes X | 4 \rangle = \langle 4 | 4 \rangle = -1$

